LAYAN KAUSHIK

Duke University
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EDUCATION

Duke University, Durham, NC, USA *Aug 2022 – May 2024 (Expected)*
M.S. Economics and Computation **GPA: 3.95/4.0**
Courses: Theory and Algorithm of Machine Learning, Data Science, Elements of Deep Learning, Advanced Algorithms, Advanced Econometrics, Numerical Analysis, Graph Network Analysis.

Indian Institute of Technology, Roorkee, India *Jul 2016 – May 2020*
B.Tech. Civil Engineering, First Division with Distinction **GPA: 8.726/10**
Ranked 12 in a class of 121 students.
Courses: Computer Programming and Numerical Analysis, Probability and Statistics, Data Mining, Linear Optimization, Graduate Econometrics.

HONORS AND AWARDS

Dean's Research Award for Master's Students, Duke University *2023*
U.S. Army Corps of Engineers Institute for Water Resources Research Grant, Washington DC *2023*
Duke Graduate Working Group on Global Issues Award, Duke University *2023*
Duke Economics Master's Scholar Award, Duke University *2022-24*
K. C. Mahindra Scholarship for Post Graduate Studies Abroad, K. C. Mahindra *2022*
Annual Excellence Award, IIT Roorkee Heritage Foundation *2018* & *2019*
Chief Minister's Scholarship, Government of Assam, Assam, India *2018*
IASc-INSA-NASI Summer Research Fellowship, Indian Academy of Sciences *2018*
IIT Roorkee ENCORE Award, IIT Roorkee *2017*
NEC Merit Scholarship, Directorate of Technical Education, Government of Assam *2016*
Top 3 in State Mathematics Olympiad, Homi Bhabha Centre for Science Education, India *2014*

WORK EXPERIENCE

Associate Analyst & Consultant, Mastercard, India *Aug 2020 - Jun 2022*

- Worked as a Data Analyst and as a Data Lead within Mastercard's Geographic Insight platform, ensuring data accuracy and generating key geographic insights.

Field Engineer Trainee, Schlumberger (Rebranded as SLB), India *May 2019 - Jul 2019*

- Worked as a Field Engineer Trainee in the upstream Oil & Gas division.

RESEARCH PROJECTS

Treatment Effect Estimation With Measurement Error in Covariates and Repeat Measurements, Guided by Prof. Michael Pollmann, Duke University *May 2023 - Present*

- Derived and implemented an unbiased objective method for higher-order polynomials under homoskedastic measurement error and multiple mismeasured variables from "Identification and Estimation of Polynomial Errors-in-Variables Models" by Hausman et al.
- Collaboratively developed an R package, optimizing existing codes for time and space efficiency, and executed extensive large dataset simulations on Duke's Computer Cluster.
- Implemented Kenichi Nagasawa's "Treatment Effect Estimation with Noisy Conditioning Variables" in R, demonstrating the model's versatility across various datasets.
- Crafted and executed cross-validation code in R for CoCoLasso regularization in linear regression and linear inverse probability weighted estimation, addressing datasets with measurement errors.

- Skills: R, C++, Causal Inference, Machine Learning

Data Enhancement and Machine Learning for Enhanced Flood Damage Assessment, Guided by Prof. Mark Borsuk, Duke University ***May 2023 - Present***

- Addressed dataset inconsistencies, including missing data and measurement discrepancies, to enhance the reliability of the Flood Insurance Claims dataset.
- Applied advanced unsupervised and semi-supervised outlier detection methods (using scikit-learn, PyOD) to discern between inches and feet measurements in the water-depth variable.
- Utilized the expectation-maximization algorithm for imputing missing values in key variables, ensuring a comprehensive dataset.
- Enhancing flood risk assessment through machine learning, comparing leading models with USACE depth damage curves, and introducing a novel hybrid approach that merges empirical data with expert insights.

Spatial Distribution of Moisture Content in Cement Mortar Structures using Radial Basis Functions, Guided by Prof. Bishwajit Bhattacharjee, IIT Delhi ***May 2018 - Aug 2018***

- Conducted extensive research on structural engineering, sustainable environment, basic electrical science, and interpolation methods to measure moisture content in cement mortar required to obtain durability and service life of the structure.
- Incorporated an efficient, accurate, and reliable method to estimate the moisture content in laboratory-synthesized cement concrete blocks using data obtained from experiments conducted during the research.
- Employed interpolation techniques like the Gaussian and inverse multiquadric radial basis function method in the estimation function to determine moisture content for which data could not be generated and verified with the theoretical calculations.
- Skills: Python (Libraries: NumPy, Matplotlib, SciPy).

ACADEMIC PROJECTS

Graph-Based Anomaly Detection using Spatial Information, Guided by Prof. Xiaobai Sun, Duke University ***Sep 2023 - Present***

- Working on designing a novel algorithm combining graph-based robust anomaly detection using location data and feature engineering.

Ride Sharing and Tipping Behavior in Chicago 2018-2023, Guided by Prof. Jian Pei, Duke University ***Jan 2023 - May 2023***

- Analyzed 300M+ Chicago ride-sharing records (2018-2023) using regression, time-series analysis, and hypothesis tests; identified statistically significant patterns, including a 33% ride drop post-pandemic and 1.5% tipping increment, reflecting socio-economic trends.
- Utilized Python for large dataset management; implemented chunk-based data processing, batch processing, and dynamic sampling to optimize data tasks.
- Skills: Python, R, Time-Series Analysis, Regression, Hypothesis Testing.

Uber Movement Data replication and Here Maps API simulations, Guided by Prof. Amit Agarwal, IIT Roorkee ***Jan 2019 - May 2019***

- Extracted real-time data from HERE MAPS API and successfully compared it with the performance indicators of Uber Movement data of New Delhi to detect patterns and analyze the impact of major events like national holidays, festivals, and rush hours on traffic conditions.
- Replicated the indicators of Uber Movement data using open-sourced data.
- Skills: Python (Libraries: osmnx, pandas, NumPy, Matplotlib, geopandas).

TEACHING EXPERIENCE

Graduate TA, CS230: Discrete Math for CS, Duke University *Jun 2023 - Aug 2023*
Responsibilities include conducting office hours, developing recitation slides, and delivering recitation lectures.

Graduate TA, CS216: Everything Data, Duke University *Jan 2023 - May 2023*
Responsibilities include holding office hours, managing undergraduate TAs, creating assignments, and automating grading.

Undergraduate TA, Computer Programming in C++, IIT Roorkee *Jan 2018- May 2018*
Held office hours and lab sessions with over 40 students.

Undergraduate TA, Solid Mechanics, IIT Roorkee *Jul 2017 - Nov 2017*
Held office hours and tutorial sessions with over 40 students.

SKILLS

Programming Languages: Python, R, SQL, C++, Matlab, LaTeX
ML Frameworks: PyTorch, JAX
Software Packages: Stata, Tableau, MS Excel, AutoCAD
Spoken Languages: English (fluent), Assamese (native speaker), Hindi (fluent)

VOLUNTEER WORK AND OTHER ACTIVITIES

Assistant Badminton Coach, Duke University *Aug 2022 - May 2023*
Provide regular training and coaching advice to players affiliated with Duke Badminton Club.

Student Mentor, Student Mentorship Program, IIT Roorkee *Jul 2018 - May 2019*
Mentored a batch of 7 freshmen to help with their transition from high school to university.

Professional Player Badminton, India *Jan 2010 - May 2020*
Represented the Indian state of Assam in various Sub-Junior and Junior National Championships in India. Was awarded Best Badminton Player of IIT Roorkee for three consecutive years 2017, 2018, and 2019.